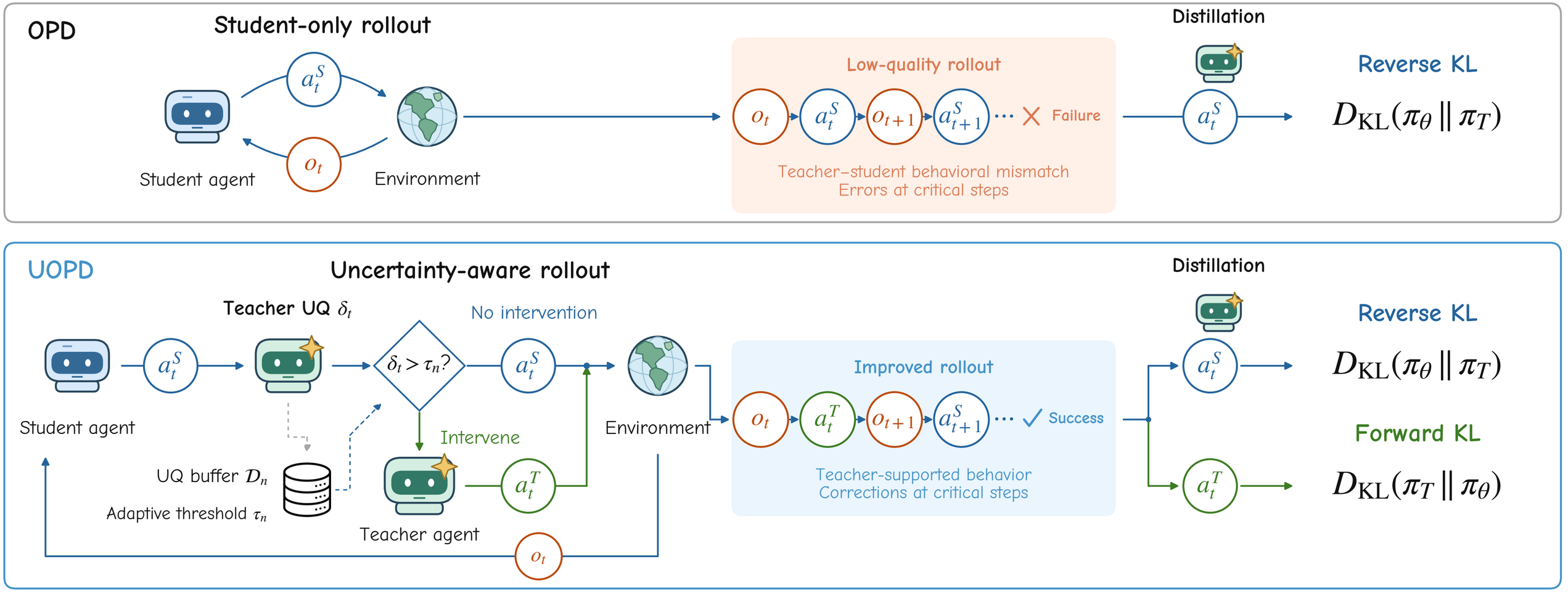
OPD
Student-only rollout
a_t^S
o_t
Student agent
Environment
Low-quality rollout
o_t a_t^S o_{t+1} a_{t+1}^S ⋯ Failure
Teacher–student behavioral mismatch
Errors at critical steps
Distillation
a_t^S
Reverse KL
$D_{\mathrm{KL}}(\pi_\theta \,\|\, \pi_T)$
UOPD
Uncertainty-aware rollout
Teacher UQ δ_t
No intervention
$\delta_t > \tau_n$?
a_t^S
Student agent
Intervene
a_t^T
Teacher agent
UQ buffer $\mathcal{D}_n$
Adaptive threshold τ_n
o_t
Environment
Improved rollout
o_t a_t^T o_{t+1} a_{t+1}^S ⋯ Success
Teacher-supported behavior
Corrections at critical steps
Distillation
a_t^S
Reverse KL
$D_{\mathrm{KL}}(\pi_\theta \,\|\, \pi_T)$
a_t^T
Forward KL
$D_{\mathrm{KL}}(\pi_T \,\|\, \pi_\theta)$